



PTT Global Chemical Public Company Limited

GC Management System and Process Safety

W-(Q-MP)-002

Operational Risk Assessment Matrix and Its Applications

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Related Units

Unit Code	Unit Name
Q-MP	GC Management System and Process Safety
T-TE	Technology
TP-RE	Reliability and Engineering
Q-MP-TS	Technical Safety Service
Q-MP-IO	Incident Investigation and ORM
S-RC-RM	Corporate Risk Management
Q-SH-PO	SHE - Polymers
TP-PM	Project Management Office
Q-MP-SS	Safety Strategy and Policy
Q-EH-ES	Environment Strategy & Policy

Unit Code	Unit Name
O-MN2-RA	Reliability and Asset Integrity

Related KPI

KPI Measure	Description / Calculation	Target (unit)

Related Law

Law Name

Related Documents

Document ID	Document Name
P-(Q-MP)-OEMS-004	Incident Investigation System
P-(Q-MP)-OEMS-005	Hazard and Operability Study (HAZOP)
P-(TP-RE)-001	Safety Instrumented Function (SIF) System Management
P-(TP-RE)-012	Risk Based Inspection (RBI) System Management
P-(TP-RE)-013	Reliability Centered Maintenance (RCM) System Management Procedure
P-(TP-RE)-OEMS-001	Operational Technology (OT) Cyber Security Management Procedure
P-(TP-RE)-OEMS-017	Computerized Maintenance Management System(CMMS)
P-(TP-RE)-OEMS-019	Maintenance Work Planning, Scheduling and Execution
P-(TP-RE)-OEMS-022	Maintenance Long Range Plan (MLRP)
P-(T-TE)-OEMS-003	Operating Window Procedure
SG-(TP-RE)-002	Functional Safety Analysis (FSA) Guidance for SIF study

External Reference Documents

Document Name

Internal Use Only



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1. Purpose/Objective

PTTGC is fully committed to ensuring a safe operating environment. A comprehensive risk assessment can reduce integrity, safety, health and environmental risk and allow us to make informed decisions on the best way to mitigate risks. **PTTGC** understands the important of risk assessment, control and mitigation, so the Risk Assessment Matrix (**RAM**) has been established and widely implemented across the company to enable a consistent approach to qualitative risk assessment.

The objectives of this **RAM** guide are:

- To achieve a consistent approach in the application of the **RAM**.
- To support the application of the **RAM** in the assessment for risks and prioritization for issues.
- To enable users to understand the concepts of qualitative risk assessment and the **RAM**.



2. Scope

Application of the **RAM** can be used either to support judgments about the significance of risks or to prioritize activities or allocate resources.

PTTGC intends that risk assessments shall be conducted for the following processes:-

- Process Hazard Analysis (**PHA** techniques: HAZOP, Bow Tie, What-if etc.)
- Justification for Project Category SHE (Safety, Health and Environment) & Law
- Incident Investigation Severity
- Risk and Reliability Management (**RRM: RCM / RBI / SIF** etc.)
- Corrective Maintenance Work Priority
- OT Cyber Security Risk Assessment
- RAM for IWO Risk Assessment
- RAM for Relations of DIW and PHA of GC Risk Assessment Matrix
- RAM for Maintenance CAPEX Decision

Thus, the **RAM** has been tailored for each process as following detail in this document.

The **RAMs** in this document are defined to be the Operational **RAM** that does not cover Corporate Risk which will be managed with Corporate Risk Management process by Corporate Risk Management (S-RC-RM).



3. Roles and Responsibility

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4. Workflow

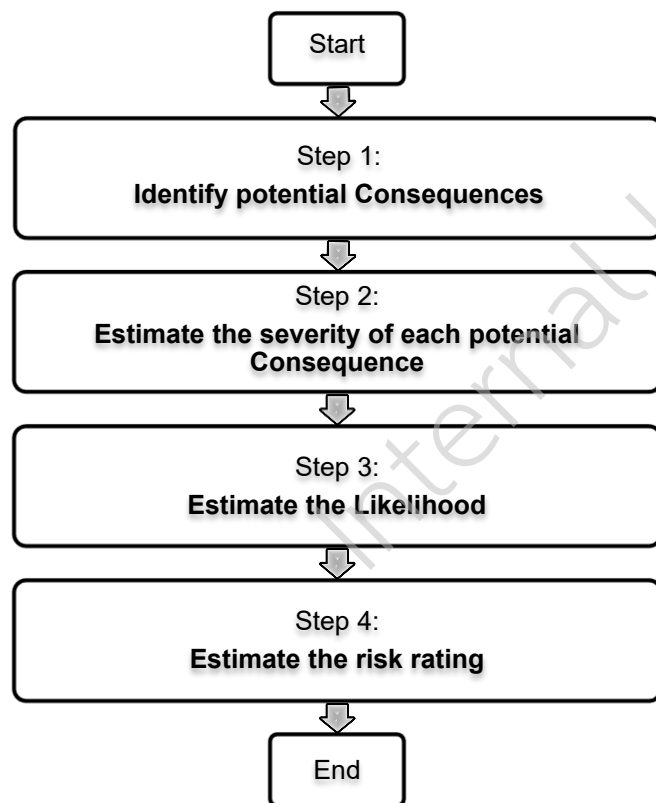


Fig. 1 General assessment process of using *RAM*



5. Detailed Narrative of Workflow

5.1 Instructions for use of the RAM

The starting point for **RAM** assessment is an understanding of the particular being considered such as Hazard or Failure in its context (activity, location etc.). Since **RAM** has been tailor made for many application as detailed in clause 8.2.1.3, it is essential to follow the specific guidance for each such process. A general assessment process consists of the following steps.

Step 1: Identify potential Consequences

Identify the Consequences that could develop from the particular being considered under the prevailing conditions. Ask the question “What ifs”, however, don’t be too imaginative:

- Avoid “what ifs” that could not occur under the circumstances of the particular being considered.
- Avoid highly unlikely “what ifs”.
- Avoid stringing too many “what ifs” together to produce an incredible Consequence.

Step 2: Estimate the Severity of each potential Consequence

For each of the identified Consequences assess the severity (1-5) in the four Consequence Categories - PEES. The severities for the PEES categories are defined in clause 8.2.1.2.2.

Step 3: Estimate the Likelihood



For each of the potential Consequences make an estimate of the Likelihood of the Consequence in terms of the Likelihood levels (1) to (5).

The Likelihood level may be judged from past experience. Try asking the question such as: “How often in the past has a hazard release resulted in a Consequence similar to the one that we are considering?” The approach is one of applying history to predict the future.

The estimate of Likelihood should be based on the particular Consequence under consideration. Not on the Likelihood of the particular being considered, the Hazard being realized or incident occurring.

The reliability of the Likelihood estimate, and therefore of the **RAM** assessment, depends to a large extent on the availability of data on previous incidents or equipment failure history and on the knowledge and experience of the assessors. It is therefore important to maintain databases and history of relevant data and make them available to everyone who will be making **RAM** assessments.

The hazard release scenario, the equipment failure history or the incident under consideration will often not be identical to the previous cases that are being used to predict Likelihood. Also, detailed information on previous case outside the organization, or even outside the Location, may not be readily available. Therefore, a combination of available information and judgment from experience has to be applied to make a best estimate of the Likelihood level (1) to (5).



Step 4: Estimate the risk rating

For each potential Consequence determine the risk rating or criticality for each of the applicable PEES categories in terms of the Consequence Severity and the Likelihood. The risk ratings can be plotted on the matrix to provide a visual representation of the risk profile of the particular being considered.

RAM is used in a number of processes for specific application. Each of these processes is governed by its own detailed instruction, which specify how the **RAM** is to be used in the specific process. Thus it is essential to consult the detailed guidance or related procedure and not to rely only on this document.



6. Appendix

6.1 Terms and Definitions

PTTGC	PTT Global Chemical Public Co., Ltd. and subsidiaries
RAM	Risk Assessment Matrix for Operational risk assessment (does not cover Corporate Risk)
Master RAM	The master template of RAM with the master Consequence and Likelihood and its description.
Application RAM	The RAM for specific application that was tailored from Master RAM .
RRM	Risk and Reliability Management
RCM	Reliability Centered Maintenance
RBI	Risk Base Inspection
SIF	Safety Instrumented Function
PHA	Process Hazard Analysis
HAZOP	Hazard and Operability Study: a structured and systematic technique for examining a defined system, with the objective of identifying potential hazards and operability problems in the system.
The Environment	Based on the workshop conducted amongst PTT Group-wide stakeholders, the following specifications are excluded from reporting: • Spill or parts of spills to secondary containment or other impermeable surfaces that do not reach the environment is excluded. • Earthen bunds are not counted as secondary containment unless they are engineered to be sufficiently impervious to prevent spill from contaminating underlying soil and/or groundwater.



- Spill from non-operated joint ventures is excluded.
- For Trading business, (since its business nature without production process i.e. oil terminal, supply & logistics and retail businesses) is not included in the scope of reporting.

Sensitive area	Area which contains natural resources, special environment, sensitive environment or area which have potential to change in negative area when receives the direct or indirect effect from environment such as watershed area type 1, protected area, biodiversity 'hotspot', habitat of critically endangered species, river, wetland, glacier or other fresh water reserve used for water supply, arctic environment with slow regeneration/turnover rates, etc. (Reference: the Notification of Ministry of Energy, Re: Prescribing Type and Size of Natural Gas Pipeline Transportation System, and Criteria, Method, and Procedure for the Environmental Report Preparation, 2013)
Chemical	Hazardous substances which are explosive, inflammable substance, oxidizing agent and peroxide substance, toxic substance, infectious substance, radioactive substance, mutagen, corrosive substance, irritating substance, other substances whether chemical or else, which may be harmful to person, animal, plant, property or environment. (Reference: Hazardous Substance Act, B.E. 2535 (1992))
Hydrocarbon	Chemical and oil that the main elements consist of hydrogen and carbon, which impact to environment.
Non-hydrocarbon	Chemical and oil that are not consisted of hydrogen and carbon, which impact to environment. (Reference: www.clas.ufl.edu , http://chemistry.tutorvista.com)



6.2 Other Supporting Information

6.2.1 RAM and Its applications

6.2.1.1 Description of **RAM**

The **RAM** is a tool to perform risk assessments. Risks are assessed on severity of Consequence and on its Likelihood. The consideration of Likelihood and Consequences leads to a rating (or ranking) of the risk, which can assist evaluation of the best method of handling the risk.

The Consequence severities scales (1-5) for each PEES (People, Environment, Economic and Social) category are not to be inferred as equal. In particular, they are not to be used to infer the value of a life. **RAM** considers the impact or Consequence on:-

- People (Safety, Health and Morale)
- Environment
- Economic (Total loss)
- Social (Community, Reputation, Customer and Law/Regulation)

For more detail of these Consequences are explained in clause 6.2.1.2.2.

The Likelihood levels (1) – (5) are defined in clause 6.2.1.2.3.

6.2.1.2 Master RAM



6.2.1.2.1 Master Template

Likelihood	General Term (No.)	Specific Application Term	Severity / Criticality / Priority Rating				
	Frequent (5)	Depend on application	Low	Medium	High	Extreme	Extreme
	Likely (4)	Depend on application	Low	Medium	High	High	Extreme
	Possible (3)	Depend on application	Low	Low	Medium	High	High
	Unlikely (2)	Depend on application	Very Low	Low	Low	Medium	Medium
	Improbable (1)	Depend on application	Very Low	Very Low	Low	Low	Low
Consequence	Severity Number		(1)	(2)	(3)	(4)	(5)
	People (Safety, Health, Morale)		- No injury or First Aid case - No or very low health effect - No or Minimal morale impact	- Medical treatment or Restrict to work - Low health effect - Short-term morale impact	- Loss time injury - Medium health effect - Long-term morale impact	- Single fatality or Permanent total disabilities - High health effect - Protesters rally or official complaint	- More than one fatality - High health effect - Employees or Contractors strike



Environment		No/Slight Effect	Minor Effect	Moderate Effect	Major Effect	Massive Effect
Economic (Total Loss)	Upstream Plant (THB)	< 0.3 M	0.3 - < 3 M	3 - < 30 M	30 - < 300 M	≥ 300 M
	Downstream Plant (THB)	< 0.1 M	0.1 - < 1 M	1 - < 10 M	10 - < 100 M	≥ 100 M
	GC-S and smaller Plant (THB)	< 0.01 M	0.01 - < 0.1 M	0.1 - < 1 M	1 - < 10 M	≥ 10 M
Social (Community, Reputation, Customer, Law/Regulation)		- No or Slight impact to Community, Reputation and Customer - No fault or insignificant fault of complying with laws/articles of association.	- Minor impact with short term recovery - Local media - Verbal complaints - Partly comply with laws/articles of association.	- Moderate impact with long term recovery - Regional media - Official letter complaint - Non-compliance with laws/articles of association	- Major impact with national concern - National media - Customer less purchase - Violate the laws/ articles of association.	- Massive impact with international concern - International media - Customer stop purchase - Violate the laws/ articles of association, and/or subject to order to dissolve the company



6.2.1.2.2 Consequence Categories and Severities

The following tables contain the description of the severity levels in each of the PEES (People, Environment, Economic, and Social) categories, followed by criteria or examples as bullet points.

Consequence on **People** includes the effect to Safety, Health and Morale.

Level (Scaling no.)	Definition
Very low (1)	No /Slight effect – not affecting work performance and not affecting daily life activities, <i>Examples:</i> • No injury or First aid case. • No or Very low health effect; ○ exposure to health hazards that give rise to noticeable discomfort, minor irritation or transient effects reversible after exposure stops. • No or Minimal morale impact.
Low (2)	Minor effect – affecting work performance in the short term, <i>Examples:</i> • Medical treatment or Restriction to work. • Low health effect; ○ illnesses such as skin irritation or food poisoning. • A minor unplanned or uncontrolled release (loss of primary containment : LOPC) of hazardous material, from a process that results a <u>release less than</u> the threshold quantities described in Table 2 of “ANSI/API RECOMMENDED PRACTICE 754” in any one-hour period. • A small fire or explosion. • Short-term morale impact.
Medium (3)	Moderate effect – affecting work performance in the longer term, <i>Examples:</i> • Loss time injury.



Consequence on **People** includes the effect to Safety, Health and Morale.

Level (Scaling no.)	Definition
	• Long term disabilities. • Medium health effect; ○ illnesses such as sensitization, noise induced hearing loss, chronic back injury, repetitive strain injury or stress. • An unplanned or uncontrolled release (loss of primary containment : LOPC) of hazardous material, from a process that results <u>a release greater than the threshold quantities described in Table 2 (but less than threshold in Table 1) of “ANSI/API RECOMMENDED PRACTICE 754”</u> in any one-hour period. • Long-term morale impact.
High (4)	Major effect – resulting from injury, occupational illness or morale, <i>Examples:</i> • Single fatality or Permanent total disabilities. • High health effect; ○ illnesses such as corrosive burns, asbestosis, silicosis, cancer and serious work related depression. • An unplanned or uncontrolled release (loss of primary containment : LOPC) of hazardous material, from a process that results <u>a release greater than the threshold quantities described in Table 1 of “ANSI/API RECOMMENDED PRACTICE 754”</u> in any one-hour period. • Protesters rally or official complaint.
Extreme (5)	Massive effect – resulting from injury, occupational illness or morale, <i>Examples:</i> • More than one fatality • Major fire or explosion resulting in more than 1 fatalities. • Very high health effect.



Consequence on **People** includes the effect to Safety, Health and Morale.

Level (Scaling no.)	Definition
	● Multiple asbestosis cases traced to a single exposure situation. ○ Cancer to a large exposed population. ● Employees or Contractors strike.

Note:

See 6.2.2 for part of Table 1 and Table 2 of “ANSI/API RECOMMENDED PRACTICE 754”. Anyhow for more and most updated details of hazardous material release from a primary containment of process, “ANSI/API RECOMMENDED PRACTICE 754 - Process Safety Performance Indicators for the Refining and Petrochemical Industries” shall be referred.

Environmental Consequence:

The bullet points in the environmental effect table are a mixture of:

- Effects, e.g. groundwater contamination.
- Events with the potential for environmental effect, e.g. exceeding a limit.
- Indicators of potential effects, e.g. complaints.

Level (Scaling no.)	Definition
Very low (1)	No/Slight effect: Minor environmental damage – contained within the premises. <u>Example:</u> ● Small spill in process area or tank farm area that readily evaporates.



Low (2)	Minor effect: Minor environmental damage, but no lasting effect. <u>Example:</u> ● Small spill off-site that seeps into the ground. ● On-site groundwater contamination. ● Complaints from community up to 10 individuals in the same case. ● Single exceeding of statutory or other prescribed limit. ● Spill any loss of containment of hydrocarbon and non-hydrocarbon of not more than 1 barrel or 159 Liters that reaches the environment, irrespective of quantity recovered.
Medium (3)	Moderate effect: Limited environmental damage that will persist or require cleaning up. <u>Example:</u> ● Spill into soil/sand that requires removal and disposal of a large quantity of soil/sand. ● Observed off-site effects or damage, e.g. fish kill or damaged vegetation. ● Off-site groundwater contamination. ● Complaints from community organizations (or more than 10 individuals in the same case). ● Frequent exceeding of statutory or other prescribed limit, with potential long term effect. ● Spill any loss of containment of hydrocarbon and non-hydrocarbon of greater than 1 barrel or 159 Liters that reaches the environment, irrespective of quantity recovered.
High (4)	Major effect: Severe environmental damage that will require extensive measures to restore beneficial uses of the environment. <u>Example:</u> ● Oil spill at a jetty during tanker (Off) loading that ends up on local beaches, requiring clean-up operations. ● Off-site groundwater contamination over an extensive area. ● Many complaints from community organization or local authorities/government. ● Extended exceeding of statutory or other prescribed limits, with potential long term effects. ● Spill any loss of containment of hydrocarbon and non-hydrocarbon of greater than 50 barrel or 7950 Liters that reaches the environment, irrespective of quantity recovered.



Extreme (5)	Massive effect: Persistent severe environmental damage that will lead to loss of commercial, recreational use or loss of natural resources over a wide area. <u>Example:</u> ● Spillage resulting in pollution of a large part of a river estuary/the sea and extensive clean-up and remediation measures. ● Significant spill: - Any hydrocarbon or Any non-hydrocarbon spill of greater than 100 barrels or 15900 Liters released into the environment; - Any hydrocarbon or Any non-hydrocarbon spill released into a sensitive environment - Any hydrocarbon or Any non-hydrocarbon spill is reported in the organization's financial statement. (1 barrel = 159 Liters, Based on assumption that barrel of oil and chemicals is the same quantity.)
Economic Consequence is the total financial loss that includes: ● Product loss (It consider Product Loss, Flare Loss or Product Off-spec, which Product price consider as average last year Price of Product) ● Opportunity loss (It consider Margins Loss , which Margins price consider as Margin yearly plan) ● Asset damage loss ● Clean up Cost	
Level (Scaling no.)	Definition
Very low (1)	No/Slight damage ● Upstream Plant: Costs < 0.3 M THB ● Downstream Plant: Costs < 0.1 M THB ● GC-S and smaller Plant: Costs < 0.01 M THB
Low (2)	Minor damage ● Upstream Plant: Costs: Costs 0.3 - <3 M THB ● Downstream Plant: Costs 0.1 - <1 M THB.



	GC-S and smaller Plant: Costs 0.01 - < 0.1 M THB
Medium (3)	Moderate damage - Upstream Plant: Costs: Costs 3 - <30 M THB - Downstream Plant: Costs 1 - <10 M THB. - GC-S and smaller Plant: Costs 0.1 - < 1 M THB
High (4)	Major damage - Upstream Plant: Costs: Costs 30 - <300 M THB - Downstream Plant: Costs 10 - <100 M THB. - GC-S and smaller Plant: Costs 1 - < 10 M THB
Extreme (5)	Massive damage - Upstream Plant: Costs: Costs $\geq$ 300 M THB - Downstream Plant: Costs $\geq$ 100 M THB. - GC-S and smaller Plant: Costs $\geq$ 10 M THB

Note:

- Upstream Plant : O-P1, O-P2, O-P3, O-P4, U-P1, U-CM, R-P1, R-RM, A-P1, A-P2
- Downstream Plant : P-HD1, P-HD1, P-LD1, P-LL1, P-LL2, PH-P1, PH-P2, EO/EG, EA, GCO, GCP, GGC, Vencorex
- GC-S and smaller Plant : GCS

Social Consequence includes the effect to Community, Reputation, Customer and Law & Regulation

Level (Scaling no.)	Definition
Very low (1)	No/Slight impact



***Social** Consequence includes the effect to Community, Reputation, Customer and Law & Regulation*

Level (Scaling no.)	Definition
	● Local public awareness but no discernible concern. ● No Media coverage. ● No/Slight impact to customer. ● No fault or insignificant fault (of complying with laws/articles of association) though within acceptable circumstances.
Low (2)	Minor impact ● Minor impact to the community with short term recovery. ● Normal community functioning with some inconvenience. ● Local public concern. ● Local media coverage. ● Verbal complaints by customer ● Partly comply with laws/articles of association, however, the contradiction are within manageable circumstances and/or may be subject to warning/fine
Medium (3)	Moderate impact – Significant impact in region or country ● Moderate impact to the community with long term recovery. ● Public protective measures (e.g. road closure). ● Regional public concern. ● Local stakeholders, e.g. community, NGO, industry and government, are aware. ● Extensive attention in local media. Some regional media coverage. ● Official letter complaint from customer. ● Non-compliance with laws/articles of association and being brought action against.



***Social** Consequence includes the effect to Community, Reputation, Customer and Law & Regulation*

Level (Scaling no.)	Definition
High (4)	Major impact – likely to escalate and affect Group reputation ● An officially declared community evacuation or community shelter-in-place. ● Major impact with national public concern. ● Impact on local and national stakeholder relations. National government and NGO involvement with potential for international NGO action. ● Extensive attention in national media. Some international coverage. ● Customer less purchase. ● Violate the laws/ articles of association and/or subject to temporary suspension order to operate.
Extreme (5)	Massive impact – Severe impact on Group reputation ● Massive impact with International public concern. ● Serious damage to infrastructure causing significant disruption to, or loss of, key services for prolonged period. Community unable to function without significant support. ● High level of concern amongst governments and action by international NGOs. ● International media attention. ● Customer stops purchase. ● Violate the laws/ articles of association, subject to suspension order to operate until further decree otherwise by court and/or subject to order to dissolve the company.



6.2.1.2.3 Likelihood Scale

The scale of increasing Likelihood is intended to represent a range from “Improbable” (highly unlikely) to “Frequent”. These terms should be used as general term on Likelihood in every application of the **RAM** so as to promote consistent assessment of risk. For each specific **Application RAM**, it will be provided “Specific Term” of Likelihood with detailed guidance for applying of those terms.

		Increasing Likelihood				
Level (Number)		(1)	(2)	(3)	(4)	(5)
General Term		Improbable	Unlikely	Possible	Likely	Frequent
Application	Specific Term	Depend on application	Depend on application	Depend on application	Depend on application	Depend on application
● PHA ● Justification for Project Category SHE & Law ● Incident Investigation	How often?	Unlikely to occur in the Industry (or has not occurred)	Possible to occur in the Industry (or has occurred)	Has happened in the PTTGC group or more than once / year in the Industry	Has happened at the Location or more than once per year in the PTTGC group	Has happened more than once per year in the Location
● RCM	Occurrence X Detection	1 - 20	21 - 40	41 - 60	61 - 80	81 - 100
● RBI	Technical Module Subfactor (TMSF)	< 1	1 - 10	10 - 100	100 - 1000	> 1000
● SIF	Demand Rate	> 1000 yr	100 - 1000 yr	10 - 100 yr	1 - 10 yr	0 - 1 yr



● CM Work Priority	Estimate Time To Failure	> 3 months	1 month – 3 months	2 weeks – 1 month	2 days – 2 weeks	Already failed or < 2 days
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The terms industry, Organization and Location are defined as;

- Industry = Refinery or Petrochemical or Chemical
- Location = Production Unit Area or Operating Area



6.2.1.3 Application RAMs

6.2.1.3.1 RAM for Process Hazard Analysis (PHA)

Likelihood	General Term (No.)	How often?	Hazard Severity Rating				
	Frequent (5)	Has happened more than once per year in the Location	Low	Medium	High	Extreme	Extreme
	Likely (4)	Has happened at the Location or more than once per year in the PTTGC group	Low	Medium	High	High	Extreme
	Possible (3)	Has happened in the PTTGC group or more than once / year in the Industry	Low	Low	Medium	High	High
	Unlikely (2)	Possible to occur in the Industry (or has occurred)	Very Low	Low	Low	Medium	Medium
	Improbable (1)	Unlikely to occur in the Industry (or has not occurred)	Very Low	Very Low	Low	Low	Low
Consequence	Severity Number		(1)	(2)	(3)	(4)	(5)
	People (Safety, Health, Morale)		- No injury or First Aid case - No or very low health effect - No or Minimal morale impact	- Medical treatment or Restrict to work - Low health effect - Short-term morale impact	- Loss time injury - Medium health effect - Long-term morale impact	- Single fatality or Permanent total disabilities - High health effect - Protesters rally or official complaint	- More than one fatality - High health effect - Employees or Contractors strike



		Environment	No/Slight Effect	Minor Effect	Moderate Effect	Major Effect	Massive Effect
Economic (Total Loss)		Upstream Plant (THB)	< 0.3 M	0.3 - < 3 M	3 - < 30 M	30 - < 300 M	≥ 300 M
		Downstream Plant (THB)	< 0.1 M	0.1 - < 1 M	1 - < 10 M	10 - < 100 M	≥ 100 M
		GC-S and smaller Plant (THB)	< 0.01 M	0.01 - < 0.1 M	0.1 - < 1 M	1 - < 10 M	≥ 10 M
		Social (Community, Reputation, Customer, Law/Regulation)	- No or Slight impact to Community, Reputation and Customer - No fault or insignificant fault of complying with laws/articles of association.	- Minor impact with short term recovery - Local media - Verbal complaints - Partly comply with laws/articles of association.	- Moderate impact with long term recovery - Regional media - Official letter complaint - Non-compliance with laws/articles of association	- Major impact with national concern - National media - Customer less purchase - Violate the laws/ articles of association.	- Massive impact with international concern - International media - Customer stop purchase - Violate the laws/ articles of association, and/or subject to order to dissolve the company

Hazard Severity and Action

Severity Ranking is classified as follows, for more details please refer to **PHA** techniques procedure such as P-(Q-MP)-OEMS-005: Hazard and Operability Study (HAZOP) and etc.

Severity	Action Required
Extreme	Serious risk and need action & risk reduction plan immediately
High	Unacceptable risk and need action & risk reduction plan immediately
Medium	Medium risk and require risk reduction plan
Low	Acceptable risk and require review of control plan



Very low

Very low risk

Note:

For the risk assessment which is required to comply with Thai regulation, the 4x4 **RAM** shall be applied for the Process Hazard Analysis. For each of the potential Consequences shall be assessed from “(1)” to “(4)” and its Likelihood shall be leveled from “Improbable (1)” to “Likely (4)”. Accordingly, the risk or hazard severity shall be ranked from “Very Low” to “High”.



6.2.1.3.2 RAM for Justification for Project Category – SHE & Law & Not in kind Replacement

Likelihood	General Term (No.)	How often?	Criticality of Project Category – SHE & Law & Not in kind Replacement				
	Frequent (5)	Has happened more than once per year in the Location	Low	Medium	High	Extreme	Extreme
	Likely (4)	Has happened at the Location or more than once per year in the PTTGC group	Low	Medium	High	High	Extreme
	Possible (3)	Has happened in the PTTGC group or more than once / year in the Industry	Low	Low	Medium	High	High
	Unlikely (2)	Possible to occur in the Industry (or has occurred)	Very Low	Low	Low	Medium	Medium
	Improbable (1)	Unlikely to occur in the Industry (or has not occurred)	Very Low	Very Low	Low	Low	Low
Consequence	Severity Number		(1)	(2)	(3)	(4)	(5)
	People (Safety, Health, Morale)		- No injury or First Aid case - No or very low health effect - No or Minimal morale impact	- Medical treatment or Restrict to work - Low health effect - Short-term morale impact	- Loss time injury - Medium health effect - Long-term morale impact	- Single fatality or Permanent total disabilities - High health effect - Protesters rally or official complaint	- More than one fatality - High health effect - Employees or Contractors strike
	Environment		No/Slight Effect	Minor Effect	Moderate Effect	Major Effect	Massive Effect



Economic (Total Loss)		Upstream Plant (THB)	< 0.3 M	0.3 - < 3 M	3 - < 30 M	30 - < 300 M	≥ 300 M
		Downstream Plant (THB)	< 0.1 M	0.1 - < 1 M	1 - < 10 M	10 - < 100 M	≥ 100 M
		GC-S and smaller Plant (THB)	< 0.01 M	0.01 - < 0.1 M	0.1 - < 1 M	1 - < 10 M	≥ 10 M
		Social (Community, Reputation, Customer, Law/Regulation)	- No or Slight impact to Community, Reputation and Customer - No fault or insignificant fault of complying with laws/articles of association.	- Minor impact with short term recovery - Local media - Verbal complaints - Partly comply with laws/articles of association.	- Moderate impact with long term recovery - Regional media - Official letter complaint - Non-compliance with laws/articles of association	- Major impact with national concern - National media - Customer less purchase - Violate the laws/ articles of association.	- Massive impact with international concern - International media - Customer stop purchase - Violate the laws/ articles of association, and/or subject to order to dissolve the company



Criticality of Project Category SHE & Law and its Investment Justification

Criticality Ranking is classified as follows, for more details please refer to P-(TP-PM)-001

Plant Investment Management and P-(TP-PM)-002: Management of Change.

Criticality	Action Required for Investment Justification
<i>Extreme</i>	Invest to maintain the essential barriers. (ALARP, As low As Reasonably Possible) and apply J-Factor more than 0.2
<i>High</i>	
<i>Medium</i>	Apply J-factor tool for assessment, and apply J-Factor more than 0.5
<i>Low</i>	No investment.
<i>Very low</i>	



6.2.1.3.3 RAM for Incident Investigation Severity

Likelihood	General Term	How often?		Incident Severity				
	Frequent	Has happened more than once per year in the Location		Low	Medium	High	Extreme	Extreme
	Likely	Has happened at the Location <i>or</i> more than once per year in the PTTGC group		Low	Medium	High	High	Extreme
	Possible	Has happened in the PTTGC group or more than once / year in the Industry		Low	Low	Medium	High	High
	Unlikely	Possible to occur in the Industry (or has occurred)		NI	Low	Low	Medium	Medium
	Improbable	Unlikely to occur in the Industry (or has not occurred)		NI	NI	Low	Low	Low
Consequence			People (Safety, Health, Morale)	- No injury or First Aid case - Tire 3b - No or very low health effect - No or Minimal morale impact	- Medical treatment or Restrict to work - LOPC Tier 3a - Low health effect - Short-term morale impact	- Loss time injury - LOPC Tier 2 - Medium health effect - Long-term morale impact	- Single fatality or Permanent total disabilities - LOPC Tier1 - High health effect - Protesters rally or official complaint	- More than one fatality - High health effect - Employees or Contractors strike
			Environment	No/Slight Effect	Minor Effect	Moderate Effect	Major Effect	Massive Effect
	Economic (Total Loss)		Upstream Plant (THB)	< 0.3 M	0.3 - < 3 M	3 - < 30 M	30 - < 300 M	≥ 300 M
			Downstream Plant (THB)	< 0.1 M	0.1 - < 1 M	1 - < 10 M	10 - < 100 M	≥ 100 M
			GC-S and smaller Plant (THB) or Outside Restricted Plant Area (THB)	< 0.01 M	0.01 - < 0.1 M	0.1 - <1 M	1 - < 10 M	≥ 10 M



		Social (Community, Reputation, Customer, Law/Regulation)	- No or Slight impact to Community, Reputation and Customer - No fault or insignificant fault of complying with laws/articles of association.	- Minor impact with short term recovery - Local media - Verbal complaints - Partly comply with laws/articles of association.	- Moderate impact with long term recovery - Regional media - Official letter complaint - Non-compliance with laws/articles of association	- Major impact with national concern - National media - Customer less purchase - Violate the laws/ articles of association.	- Massive impact with international concern - International media - Customer stop purchase - Violate the laws/ articles of association, and/or subject to order to dissolve the company
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Note:

- A Process Safety Event per API RP 754 categorizes as Tier 1 and Tier 2 are High severity as a minimum.
- Opportunity loss (It consider Margins Loss , which Margins price consider as Margin yearly plan)
- Outside Restricted Plant Area = Area, Offices, Buildings or Facilities outside plant restricted area fence for example, such as administration building at Bangkok, fire fighting office building, maintenance workshop, warehouse (product and non-product), road and etc. which are not located inside the Restricted Plant Area.
- The risk rating of an incident is determined for Consequence and Likelihood, applying the nomal RAM method described in clause 5. The incident that is being assessed should be count in the Likelihood of the Consequence that happened in this particular incident.
- **NI** means No Investigation

Incident Severity Rating and Action

Severity Ranking is classified as follows, for more details please refer to P-(Q-MP)-OEMS-004: Incident Investigation System.



Incident Severity	Investigation Team			
	Chairman	Central Advisor	Lead Incident Investigator	Team Member
Extreme	C-Level assigned SVP	Required	Qualified Division Manager Up	Related to Incident occurred
High	VP of Incident Owner	Required	Qualified Senior Staff Up	Related to Incident occurred
Medium	DM of Incident Owner	-	Qualified Senior Staff Up	Related to Incident occurred
Low	Direct Supervisor	-	Qualified Staff	Affected personnel



6.2.1.3.4 RAM for Reliability Centered Maintenance (RCM)

Likelihood	General Term	Occurrence x Detection	RCM Criticality				
	Frequent	81 - 100	Low	Medium	High	Extreme	Extreme
	Likely	61 - 80	Low	Medium	High	High	Extreme
	Possible	41 - 60	Low	Low	Medium	High	High
	Unlikely	21 - 40	Very Low	Low	Low	Medium	Medium
	Improbable	1 - 20	Very Low	Very Low	Low	Low	Low
Consequence	People (Safety, Health)		- No injury or First Aid case - No or very low health effect	- Medical treatment or Restrict to work - Low health effect	- Loss time injury - Medium health effect	- Single fatality or Permanent total disabilities - High health effect	- More than one fatality - High health effect
	Environment		No/Slight Effect	Minor Effect	Moderate Effect	Major Effect	Massive Effect
	Economic (Total Loss)	Upstream Plant (THB)	< 0.3 M	0.3 - < 3 M	3 - < 30 M	30 - < 300 M	≥ 300 M
		Downstream Plant (THB)	< 0.1 M	0.1 - < 1 M	1 - < 10 M	10 - < 100 M	≥ 100 M
		GC-S and smaller Plant (THB)	< 0.01 M	0.01 - < 0.1 M	0.1 - < 1 M	1 - < 10 M	≥ 10 M
	Social (Community, Reputation, Customer, Law/Regulation)		- No or Slight impact to Community, Reputation and Customer - No fault or insignificant fault of complying with laws/articles of association.	- Minor impact with short term recovery - Local media - Verbal complaints - Partly comply with laws/articles of association.	- Moderate impact with long term recovery - Regional media - Official letter complaint - Non-compliance with laws/articles of association	- Major impact with national concern - National media - Customer less purchase - Violate the laws/ articles of association.	- Massive impact with international concern - International media - Customer stop purchase - Violate the laws/ articles of



					association, and/or subject to order to dissolve the company
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Note:

- Likelihood for **RCM** application shall be calculated from multiplying of Occurrence and Detection regarding to the Risk Priority Number (RPN) methodology which is a technique for analyzing the risk associated with potential problems indentified during a Failure Mode and Effects Analysis (FMEA).
 - Occurrence, which rates the Likelihood that the failure will occur.
 - Detection, which rates the Likelihood that the problem will be detected before it reaches the failed state.
- The details and their rating for calculating the **RCM** Likelihood, please refer to P-(TP-RE)-013: Reliability Centered Maintenance (RCM) System Management.



Reliability Centered Maintenance (RCM) Criticality and Action

Failure Mode Criticality ranking is classified as follows, for more details please refer to P-(TP-RE)-013: Reliability Centered Maintenance (RCM) System Management.

Failure Mode Criticality	Action Required
<i>Extreme</i>	Logic Tree and Task Selection Road Map according to SAE JA 1011 (1999-08)/ SAE JA 1012 (2002-01 A). Redesign shall be preferred if technical feasibility criteria for the Scheduled Task has not enough confidence rate.
<i>High</i>	Logic Tree and Task Selection Road Map according to SAE JA 1011 (1999-08)/ SAE JA 1012 (2002-01 A)
<i>Medium</i>	Logic Tree and Task Selection Road Map according to SAE JA 1011 (1999-08)/ SAE JA 1012 (2002-01 A)
<i>Low</i>	Non-Critical Task selection process
<i>Very low</i>	Non-Critical Task selection process



6.2.1.3.5 RAM for Risk Base Inspection (RBI)

Likelihood	General Term	Technical Module Subfactor (TMSF)		RBI Criticality				
	Frequent	> 1000		Low	Medium	High	Extreme	Extreme
	Likely	100 - 1000		Low	Medium	High	High	Extreme
	Possible	10 - 100		Low	Low	Medium	High	High
	Unlikely	1 - 10		Negligible	Low	Low	Medium	Medium
	Improbable	< 1		Negligible	Negligible	Low	Low	Low
Consequence	People (Safety, Health)	Pressure Equipment	FCC	$\leq 0 - 0.929 \text{ m}^2$	$0.929 - \leq 9.29 \text{ m}^2$	$9.29 - \leq 92.90 \text{ m}^2$	$92.9 - \leq 929 \text{ m}^2$	$> 929.0 \text{ m}^2$
			TCC	$\leq 0 - 0.929 \text{ m}^2$	$0.929 - \leq 9.29 \text{ m}^2$	$9.29 - \leq 92.90 \text{ m}^2$	$92.9 - \leq 929 \text{ m}^2$	$> 929.0 \text{ m}^2$
		Environment	No/Slight Effect	Minor Effect	Moderate Effect	Major Effect	Massive Effect	
	Economic (Total Loss)	Upstream Plant (THB)	< 0.3 M	0.3 - < 3 M	3 - < 30 M	30 - < 300 M	$\geq 300 \text{ M}$	
		Downstream Plant (THB)	< 0.1 M	0.1 - < 1 M	1 - < 10 M	10 - < 100 M	$\geq 100 \text{ M}$	
		GC-S and smaller Plant (THB)	< 0.01 M	0.01 - < 0.1 M	0.1 - <1 M	1 - < 10 M	$\geq 10 \text{ M}$	
	Social (Community, Reputation, Customer, Law/Regulation)		- No or Slight impact to Community, Reputation and Customer - No fault or insignificant fault of complying with laws/articles of association.	- Minor impact with short term recovery - Local media - Verbal complaints - Partly comply with laws/articles of association.	- Moderate impact with long term recovery - Regional media - Official letter complaint - Non-compliance with laws/articles of association	- Major impact with national concern - National media - Customer less purchase - Violate the laws/ articles of association.	- Massive impact with international concern - International media - Customer stop purchase - Violate the laws/ articles of association.	



					association, and/or subject to order to dissolve the company
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Note:

- Technical Module Subfactor (TMSF) is ratio of the frequency of failure due to damage to the generic failure frequency times the Likelihood that the damage level is present (refer to API 581).
- PSV - Pressure Safety Valve
- TCC - Toxic Consequence Category
- FCC - Flammable Consequence Category



Risk Base Inspection (RBI) Criticality and Action

Criticality Ranking is classified as follows, for more details please refer to P-(TP-RE)-004: Risk Based Inspection (RBI) System Management

Criticality	Action Required
<i>Extreme</i>	Specific risk management or Redesign
<i>High</i>	Inspection plan and Detail analysis and/or Process monitoring may be required
<i>Medium</i>	Inspection plan
<i>Low</i>	Inspection plan
<i>Negligible</i>	No additional inspection / Run to Fail methodology



6.2.1.3.6 RAM for Safety Instrumented Function (SIF)

6.2.1.3.6.1 RAM for SIF – People (Safety, Health) and Environment

Likelihood	General Term	Demand Rate	SIL				
	Frequent	<1 year	1	2	3	4	4
	Likely	1 to 10 years	1	2	3	3	4
	Possible	>10 to 100 years	A	1	2	3	3
	Unlikely	>100 to 1,000 years	N	A	1	2	2
	Improbable	>1,000 years	N	N	A	1	1
Consequence	People (Safety, Health)		- No injury or First Aid case - No or very low health effect	- Medical treatment or Restrict to work - Low health effect	- Loss time injury - Medium health effect	- Single fatality or Permanent total disabilities - High health effect	- More than one fatality - High health effect
	Environment		No/Slight Effect	Minor Effect	Moderate Effect	Major Effect	Massive Effect



6.2.1.3.6.2 **RAM for SIF – Economic**

Likelihood	General Term	Demand Rate	SIL				
	Frequent	<1 year	A	I	2	3	3
	Likely	1 to 10 years	A	I	2	2	3
	Possible	>10 to 100 years	A	A	I	2	2
	Unlikely	>100 to 1,000 years	N	A	A	I	I
	Improbable	>1,000 years	N	N	A	A	A
	Economic (Total Loss)	(THB)	< 0.3 M	0.3 - < 3 M	3 - < 30 M	30 - < 300 M	≥ 300 M

Note:

For SIF assessment, Economic consequence criteria for **Downstream Plant** and **GC-S and smaller Plant** shall use the same criteria as **Upstream Plant** since it is not economic to install complex and expensive safeguarding systems for lower value of consequence.



SIL (Safety Integrity Level) for SIF and Action

SIL (Safety Integrity Level) is classified as follows, for more details please refer to P-(TP-RE)-001: Safety Instrumented Function (SIF) System Management

SIL	Action Required
N	No <i>SIF</i> Requirement.
A	No <i>SIF</i> Requirement. Need Alarm only.
1	Follow IEC 61511
2	
3	
4	



6.2.1.3.7 RAM for Corrective Maintenance Work Priority

Likelihood	General Term	Estimate Time To Failure	Corrective Maintenance Work Priority				
	Frequent	Already failed or < 2 days	4	3	2	1	1
	Likely	2 days – 2 weeks	4	3	2	2	1
	Possible	2 weeks – 1 month	4	4	3	2	2
	Unlikely	1 month – 3 months	4	4	4	3	3
	Improbable	> 3 months	4	4	4	4	4
	Severity Level		Very Low	Low	Medium	High	Extream
Consequence	People (Safety, Health, Morale)		- No injury or First Aid case - No or very low health effect - No or Minimal morale impact	- Medical treatment or Restrict to work - Low health effect - Short-term morale impact	- Loss time injury - Medium health effect - Long-term morale impact	- Single fatality or Permanent total disabilities - High health effect - Protesters rally or official complaint	- More than one fatality - High health effect - Employees or Contractors strike
	Environment		No/Slight Effect	Minor Effect	Moderate Effect	Major Effect	Massive Effect
	Economic (Total Loss)	Upstream Plant (THB)	< 0.3 M	0.3 - < 3 M	3 - < 30 M	30 - < 300 M	≥ 300 M
		Downstream Plant (THB)	< 0.1 M	0.1 - < 1 M	1 - < 10 M	10 - < 100 M	≥ 100 M
		GC-S and smaller Plant (THB)	< 0.01 M	0.01 - < 0.1 M	0.1 - < 1 M	1 - < 10 M	≥ 10 M
	Social (Community, Reputation, Customer, Law/Regulation)		- No or Slight impact to Community, Reputation and	- Minor impact with short term recovery	- Moderate impact with long term recovery	- Major impact with national concern	- Massive impact with international concern



	Customer - No fault or insignificant fault of complying with laws/articles of association.	- Local media - Verbal complaints - Partly comply with laws/articles of association.	- Regional media - Official letter complaint - Non-compliance with laws/articles of association	- National media - Customer less purchase - Violate the laws/ articles of association.	- International media - Customer stop purchase - Violate the laws/ articles of association, and/or subject to order to dissolve the company
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Corrective Maintenance Work Priority and Action

Severity Ranking is classified as follows, for more details please refer to *P-(TP-RE)-OEMS-017: Computerized Maintenance Management System(CMMS)*

CM Priority	Action Required
1	Urgent / Call in (Attack within 1 hrs)
2	Check within 24 hrs
3	Check within 2 weeks
4	Check within 1 month



6.2.1.3.8 RAM for OT Cyber Security Risk Assessment

Likelihood	General Term	How often?	SL (Security Level)				
	Frequent	Has happened more than once per year in the Location	1	2	3	4	4
	Likely	Has happened at the Location <i>or</i> more than once per year in the PTTGC group	1	2	3	3	4
	Possible	Has happened in the PTTGC group or more than once / year in the Industry	0	1	2	3	3
	Unlikely	Possible to occur in the Industry (or has occurred)	0	0	1	2	2
	Improbable	Unlikely to occur in the Industry (or has not occurred)	0	0	0	1	1
Consequence			People (Safety, Health, Morale)	Environment			
			- No injury or First Aid case - No or very low health effect - No or Minimal morale impact	- Medical treatment or Restrict to work - Low health effect - Short-term morale impact	- Loss time injury - Medium health effect - Long-term morale impact	- Single fatality or Permanent total disabilities - High health effect - Protesters rally or official complaint	- More than one fatality - High health effect - Employees or Contractors strike
			No/Slight Effect	Minor Effect	Moderate Effect	Major Effect	Massive Effect
	Economic (Total Loss)	Upstream Plant (THB)	< 0.3 M	0.3 - < 3 M	3 - < 30 M	30 - < 300 M	≥ 300 M
		Downstream Plant (THB)	< 0.1 M	0.1 - < 1 M	1 - < 10 M	10 - < 100 M	≥ 100 M
		GC-S and smaller Plant (THB) or Outside Restricted Plant Area (THB)	< 0.01 M	0.01 - < 0.1 M	0.1 - < 1 M	1 - < 10 M	≥ 10 M



		Social (Community, Reputation, Customer, Law/Regulation)	- No or Slight impact to Community, Reputation and Customer - No fault or insignificant fault of complying with laws/articles of association.	- Minor impact with short term recovery - Local media - Verbal complaints - Partly comply with laws/articles of association.	- Moderate impact with long term recovery - Regional media - Official letter complaint - Non-compliance with laws/articles of association	- Major impact with national concern - National media - Customer less purchase - Violate the laws/ articles of association.	- Massive impact with international concern - International media - Customer stop purchase - Violate the laws/ articles of association, and/or subject to order to dissolve the company
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SL (Security Level) for OT cyber security risk assessment and Action

SL (Security Level) is classified as follows, for more details please refer to *P-(T-RE)-OEMS-001*

Operational Technology (OT) Cyber Security Management Procedure

SL	Risk Reduction Requirement
0	No specific requirements
1	Follow IEC 62443
2	
3	
4	

6.2.1.3.9 RAM for IWO Risk Assessment



Likelihood	General Term	Time to Failure	IOW Risk				
	Sudden failure (5)	$T \leq 1 \text{ Day}$	Low	Medium	High	Extreme	Extreme
	Rapid term failure (4)	$1 \text{ Day} < T \leq 3 \text{ Months}$	Low	Medium	High	High	Extreme
	Short term failure (3)	$3 \text{ Months} < T \leq 1 \text{ Year}$	Low	Low	Medium	High	High
	Middle term failure (2)	$1 \text{ Year} < T \leq 5 \text{ Years}$	Very Low	Low	Low	Medium	Medium
	Long term failure (1)	$T > 5 \text{ Years}$	Very Low	Very Low	Low	Low	Low
Consequence	Severity Number		(1)	(2)	(3)	(4)	(5)
	People (Safety, Health)		-No injury or First Aid case -No or very low health effect -No or Minimal morale impact	- Medical treatment or Restrict to work - Low health effect - Short-term morale impact	- Loss time injury - Medium health effect - Long-term morale impact	- Single fatality or Permanent total disabilities - High health effect - Protesters rally or official complaint	- More than one fatality - High health effect - Employees or Contractors strike
	Environment		No / Slight Effect	Minor Effect	Moderate Effect	Major Effect	Massive Effect
	Economic (Total Loss)	GPC (THB)	< 0.3 M	0.3 - < 3 M	3 - < 30 M	30 - < 300 M	$\geq 300 \text{ M}$
		BU (THB)	< 0.1 M	0.1 - < 1 M	1 - < 10 M	10 - < 100 M	$\geq 100 \text{ M}$
		Small BU (THB)	< 0.01 M	0.01 - < 0.1 M	0.1 - < 1 M	1 - < 10 M	$\geq 10 \text{ M}$
	Social (Community, Reputation, Customer, Law/Regulation)		- No or Slight impact to Community, Reputation and	- Minor impact with short term recovery	- Moderate impact with long term recovery	- Major impact with national concern	- Massive impact with international concern



Likelihood	General Term	Time to Failure	IOW Risk				
	Sudden failure (5)	$T \leq 1 \text{ Day}$	Low	Medium	High	Extreme	Extreme
	Rapid term failure (4)	$1 \text{ Day} < T \leq 3 \text{ Months}$	Low	Medium	High	High	Extreme
	Short term failure (3)	$3 \text{ Months} < T \leq 1 \text{ Year}$	Low	Low	Medium	High	High
	Middle term failure (2)	$1 \text{ Year} < T \leq 5 \text{ Years}$	Very Low	Low	Low	Medium	Medium
	Long term failure (1)	$T > 5 \text{ Years}$	Very Low	Very Low	Low	Low	Low
			Customer - No fault or insignificant fault of complying with laws/articles of association.	-Local media -Verbal complaints -Partly comply with laws/articles of association	- Regional media - Official letter complaint - Non-compliance with laws/articles of association	- National media - Customer less purchase - Violate the laws/ articles of association.	- International media - Customer stop purchase - Violate the laws/ articles of association, and/or subject to order to dissolve the company



SL (Security Level) for IWO risk assessment and Action

SL (Security Level) is classified as follows, for more details please refer to P-(T-TE)-OEMS-003: Operating Window Procedure

SL	Type of IOW	Action guideline
0	Information IOW I-IOW	- Parameter are tracked and trended by SME and used as information for RBI and T/A activities - Process engineer tracked and report at monthly performance report
1	Standard IOW S-IOW	- Operations and SME's are alerted/notified of exceedances - Trouble shooting initiated with planned adjustments to operation - Inspection/maintenance developed
2		
3	Critical IOW C-IOW	- Alarm and/or trip as per alarm management procedure - SME are notified Operators take urgent action to return process to normal operation
4		



6.2.1.3.10 Relations of DIW and PHA of GC Risk Assessment Matrix

For the risk assessment, which is required to comply with Thai regulation, the 4x4 RAM shall be applied.

For each of the potential Consequences shall be assessed from “(2)” to “(5)” and its Likelihood shall be leveled from “Improbable (2)” to “Likely (5)”.

Accordingly, the risk or hazard severity shall be ranked from “Low” to “Extreme”.

For economic of BU and Small BU shall be keep the values.

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Likelihood	General Term (No.)	How often?	Hazard Severity Rating			
	Frequent 5	Has happened more than once per year in the Location	Medium	High	Extreme	Extreme
	โอกาสเกิดสูง 4	ความถี่ในการเกิด เกิดมากกว่า 1 ครั้งใน 1 ปี	ความเสี่ยงที่ยอมรับได้	ความเสี่ยงสูง	ความเสี่ยงที่ยอมรับไม่ได้	ความเสี่ยงที่ยอมรับไม่ได้
	Likely 4	Has happened at the Location or more than once per year in the PTTGC group	Medium	High	High	Extreme
	โอกาสเกิดปานกลาง 3	ความถี่ในการเกิด เกิด 1 ครั้งในช่วง 1-5 ปี	ความเสี่ยงที่ยอมรับได้	ความเสี่ยงที่ยอมรับได้	ความเสี่ยงสูง	ความเสี่ยงที่ยอมรับไม่ได้
	Possible 3	Possible to happen in the PTTGC group or more than once / year in the Industry	Low	Medium	High	High
	โอกาสเกิดน้อย 2	ความถี่ในการเกิด เกิด 1 ครั้งในช่วง 5-10 ปี	ความเสี่ยงเล็กน้อย	ความเสี่ยงที่ยอมรับได้	ความเสี่ยงที่ยอมรับได้	ความเสี่ยงสูง
	Unlikely 2	Possible to occur in the Industry (or has occurred)	Low	Low	Medium	Medium
Consequence	โอกาสเกิดมาก 1	ไม่เคยเกิดเลยในช่วงเวลาตั้งแต่ 10 ปีขึ้นไป	ความเสี่ยงเล็กน้อย	ความเสี่ยงเล็กน้อย	ความเสี่ยงที่ยอมรับได้	ความเสี่ยงที่ยอมรับได้
	Severity Number		2	3	4	5
	DIW		1	2	3	4
	People (Safety, Health, Morale)		- Medical treatment or Restrict to work - Low health effect - Short-term morale impact	- Loss time injury - Medium health effect - Long-term morale impact	- Single fatality or Permanent total disabilities - High health effect - Protesters rally or official complaint	- More than one fatality - High health effect - Employees or Contractors strike
	DIW		- มีการบาดเจ็บเล็กน้อยในระดับปฐมพยาบาล	- มีการบาดเจ็บที่ต้องได้รับการรักษาทางการแพทย์	- มีการบาดเจ็บหรือเจ็บป่วยที่รุนแรง	- ทุพพลภาพ หรือเสียชีวิต
	Environment		Minor Effect	Moderate Effect	Major Effect	Massive Effect
	DIW		- มีผลกระทบต่อสิ่งแวดล้อมเล็กน้อย สามารถควบคุมแก้ไขได้	- มีผลกระทบต่อสิ่งแวดล้อมปานกลาง สามารถแก้ไขได้ในระยะเวลานั้น	- มีผลกระทบต่อสิ่งแวดล้อมรุนแรง ต้องใช้ระยะเวลานาน	- มีผลกระทบต่อสิ่งแวดล้อมรุนแรงมาก ต้องใช้ทรัพยากรและเวลานานในการแก้ไข
	Economic (Total Loss)	GPC (THB)	0.3 - < 3 M	3 - < 30 M	30 - < 300 M	≥ 300 M
		BU (THB)	< 0.1 M	0.1 - < 1 M	1 - < 10 M	10 - < 100 M
		Small BU (THB)	< 0.01 M	0.01 - < 0.1 M	0.1 - < 1 M	1 - < 10 M
	DIW		- ทรัพย์สินเสียหายน้อยมาก หรือไม่เสียหายเลย	- ทรัพย์สินเสียหายปานกลาง และสามารถดำเนินการแก้ไขได้	- ทรัพย์สินเสียหายมาก และต้องหยุดการผลิดในบางส่วน	- ทรัพย์สินเสียหายมาก และต้องหยุดการผลิดทั้งหมด
	Social		- Minor impact with short term recovery	- Moderate impact with long term recovery	- Major impact with national concern	- Massive impact with international concern
	(Community, Reputation, Customer, Law/Regulation)		- Local media	- Regional media	- National media	- International media
			- Verbal complaints	- Official letter complaint	- Customer less purchase	- Customer stop purchase
			- Partly comply with laws/articles of association.	- Non-compliance with laws/articles of association	- Violate the laws/ articles of association.	- Violate the laws/ articles of association, and/or subject to order to dissolve the company
	DIW		- ไม่มีผลกระทบต่อชุมชนรอบโรงงาน หรือมีผลกระทบเล็กน้อย	- มีผลกระทบต่อชุมชนรอบโรงงาน และแก้ไขได้ในระยะเวลานั้น	- มีผลกระทบต่อชุมชนรอบโรงงาน และต้องใช้เวลาแก้ไข	- มีผลกระทบต่อชุมชนรอบโรงงาน และต้องใช้เวลาแก้ไข



Risk Ranking

 DIW	Level 5: Extream	Serious risk and need action & risk rediuction plan immediately
	Level 4: ความเสี่ยงที่ยอมรับไม่ได้	ความเสี่ยงที่ยอมรับไม่ได้ ต้องหยุดดำเนินการและปรับปรุงแก้ไขเพื่อลดความเสี่ยงลงทันที
 DIW	Level 4: High	Unacceptable risk and need action & risk reduction plan immediately
	Level 3: ความเสี่ยงสูง	ความเสี่ยงสูง ต้องมีการดำเนินงานเพื่อลดความเสี่ยง
 DIW	Level 3: Medium	Medium risk and require reduction plan
	Level 2: ความเสี่ยงที่ยอมรับได้	ความเสี่ยงที่ยอมรับได้ ต้องมีการทบทวนมาตรการควบคุม
 DIW	Level 2: Low	Acceptable risk and require review of control plan
	Level 1: ความเสี่ยงเล็กน้อย	ความเสี่ยงเล็กน้อย

หมายเหตุ :

ผลกระทบต่อชุมชน หมายถึง เหตุรำคาญต่อชุมชน การบาดเจ็บ เจ็บป่วยของประชาชน ความเสียหายต่อทรัพย์สินของชุมชน และประชาชน

ผลกระทบต่อสิ่งแวดล้อม หมายถึง ความเสื่อมโทรม และความเสียหายของสิ่งแวดล้อม เช่น อากาศ ดิน แหล่งน้ำ เป็นต้น

ความเสียหายต่อทรัพย์สินในแต่ละระดับโรงงาน สามารถกำหนดขึ้นเองได้ตามความเหมาะสม โดยพิจารณาถึงขีดความสามารถของโรงงาน



6.2.1.3.11 RAM for Maintenance CAPEX Decision

Likelihood	General Term (No.)	Probability	Score	Example of definition	Maintenance CAPEX Decision				
	Frequent (5)	1% and less	0.8-1	Failure occur if not doing that activities, all support information shown trend that reflect the expected result surely. Past record, it used to be breakdown.	Low	Medium	High	Extreme	Extreme
	Likely (4)	>1%-30%	0.5-0.8	Trend show the expected result of failure, information is realized it will happen.	Low	Medium	High	High	Extreme
	Possible (3)	>30%-50%	0.3-0.5	It depends on other parameter that have a chance can be occurred	Low	Low	Medium	High	High
	Unlikely (2)	>50%-80%	0.1-0.3	From start up, it never happen but another similar plant has some experience, it has a chance here but seem minimal	Very Low	Low	Low	Medium	Medium
	Improbable (1)	>80%-100%	<=0.1	Never happen in the industry, unfortunately we might be the first one.	Very Low	Very Low	Low	Low	Low
	Severity Number				(1)	(2)	(3)	(4)	(5)
Consequence	People (Safety, Health, Morale)				- No injury or First Aid case - No or very low health effect - No or Minimal morale impact	- Medical treatment or Restrict to work - Low health effect - Short-term morale impact	- Loss time injury - Medium health effect - Long-term morale impact	- Single fatality or Permanent total disabilities - High health effect - Protesters rally or official complaint	- More than one fatality - High health effect - Employees or Contractors strike



Likelihood	General Term (No.)	Probability	Score	Example of definition	Maintenance CAPEX Decision				
	Frequent (5)	1% and less	0.8-1	Failure occur if not doing that activities, all support information shown trend that reflect the expected result surely. Past record, it used to be breakdown.	Low	Medium	High	Extreme	Extreme
	Likely (4)	>1%-30%	0.5-0.8	Trend show the expected result of failure, information is realized it will happen.	Low	Medium	High	High	Extreme
	Possible (3)	>30%-50%	0.3-0.5	It depends on other parameter that have a chance can be occurred	Low	Low	Medium	High	High
	Unlikely (2)	>50%-80%	0.1-0.3	From start up, it never happen but another similar plant has some experience, it has a chance here but seem minimal	Very Low	Low	Low	Medium	Medium
	Improbable (1)	>80%-100%	<=0.1	Never happen in the industry, unfortunately we might be the first one.	Very Low	Very Low	Low	Low	Low
	Environment				No/Slight Effect	Minor Effect	Moderate Effect	Major Effect	Massive Effect
	Economic (Total Loss)	Upstream Plant (THB)			< 0.3 M	0.3 - < 3 M	3 - < 30 M	30 - < 300 M	≥ 300 M
		Downstream Plant (THB)			< 0.1 M	0.1 - < 1 M	1 - < 10 M	10 - < 100 M	≥ 100 M
		GC-S and smaller Plant (THB)			< 0.01 M	0.01 - < 0.1 M	0.1 - <1 M	1 - < 10 M	≥ 10 M
Social (Community, Reputation, Customer, Law/Regulation)				- No or Slight impact to Community, Reputation and Customer	- Minor impact with short term recovery - Local media	- Moderate impact with long term recovery	- Major impact with national concern - National media	- Massive impact with international concern	



Likelihood	General Term (No.)	Probability	Score	Example of definition	Maintenance CAPEX Decision				
	Frequent (5)	1% and less	0.8-1	Failure occur if not doing that activities, all support information shown trend that reflect the expected result surely. Past record, it used to be breakdown.	Low	Medium	High	Extreme	Extreme
	Likely (4)	>1%-30%	0.5-0.8	Trend show the expected result of failure, information is realized it will happen.	Low	Medium	High	High	Extreme
	Possible (3)	>30%-50%	0.3-0.5	It depends on other parameter that have a chance can be occurred	Low	Low	Medium	High	High
	Unlikely (2)	>50%-80%	0.1-0.3	From start up, it never happen but another similar plant has some experience, it has a chance here but seem minimal	Very Low	Low	Low	Medium	Medium
	Improbable (1)	>80%-100%	<=0.1	Never happen in the industry, unfortunately we might be the first one.	Very Low	Very Low	Low	Low	Low
					- No fault or insignificant fault of complying with laws/articles of association.	- Verbal complaints - Partly comply with laws/articles of association.	- Regional media - Official letter complaint - Non-compliance with laws/articles of association	- Customer less purchase - Violate the laws/ articles of association.	- International media - Customer stop purchase - Violate the laws/ articles of association, and/or subject to order to dissolve the company



Criticality Ranking is classified as follows, for more details please refer to P-(TP-RE)-OEMS-022 Maintenance Long Range Plan (MLRP).

Criticality	Action Required for CAPEX Justification
Extreme	Invest to maintain the Safety, Reliability and Apply J-Factor greater than or equal to 1
High	
Medium	
Low	
Very low	No investment